

14. **Assertion (A):** From Thomson's experiment and Millikan's oil drop experiment, mass as well as charge of an electron can be calculated.

Reason (R): Specific charge of an electron is obtained from Thomson's experiment, whereas the charge of an electron is obtained from Millikan's oil drop experiment.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. Match the values of Column A with those of Column B.

Column A (Type of transition in Hydrogen atom)	Column B (No. of spectral lines)
(i) 5th orbit to 3rd orbit	(A) 10
(ii) 4th orbit to 1st orbit	(B) 1
(iii) 6th orbit to 2nd orbit	(C) 6
(iv) 5th orbit to 4th orbit	(D) 3

- (a) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (B); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
(b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
(c) (i) $\rightarrow$ (A); (ii) $\rightarrow$ (B); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
(d) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

Space for rough work

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. The number of electrons in the third shell of an element is seven times the number of electrons in the fourth shell. Identify the element.
(a) Iron (b) Cobalt
(c) Nickel (d) Manganese
2. Identify which of the following configurations holds good for zinc.
(a) $x, (x + y), 3y, x$
(b) $x, (x + y), 3y, (x - 1)$
(c) $x, y, (3y - 1), x$
(d) $x, x + y, (2y + 1), (x - 1)$
3. Which of the following postulates of Dalton's atomic theory became contradictory after the discovery of isotopes?
(a) Matter is made up of tiny indivisible particles called atoms.
(b) Atoms of same elements are identical in all aspects.
(c) Atoms of different elements are different in all aspects.
(d) Atom is the smallest particle or unit of matter which takes part in a chemical reaction.
4. The ratio of atomic numbers of two elements 'P' and 'Q' is 2:7. The energy of the fourth orbit of single electron species of P is -13.6 eV. Calculate the difference in the number of electrons present in valence and penultimate shells of Q.
(a) 2 (b) 4 (c) 3 (d) 1
5. Trinegative ion of element 'X' is isoelectronic with inert gas 'Y'. If X has two completely filled shells, then identify Y.
(a) Ne (b) Ar (c) Kr (d) Xe
6. According to Bohr's atomic model, _____.
(a) electrons revolve around the nucleus and continuously emit energy
(b) the energy of an electron does not alter with the change of the orbit
(c) electrons revolve around the nucleus in spiral paths called stationary orbits
(d) a fixed amount of energy is emitted when an electron jumps from an outer orbit to an inner orbit
7. α -ray scattering experiment was carried out _____.
(a) to overcome the drawbacks of Rutherford's atomic model
(b) to prove that nucleus is present at the centre of the atom
(c) to prove the correctness of J. J. Thomson's atomic model
(d) to prove the existence of positively charged particle in an atom

Space for rough work

8. The difference in weight between two isotopes of an element is 2 units and the average atomic mass is 1.25 units more than the lighter isotope. Calculate the ratio of the lighter to heavier isotope in the natural sample.
- (a) 2:1 (b) 1:2 (c) 3:5 (d) 5:3
9. If the radius of the third orbit of a single electron species is $x \text{ A}^\circ$, then find out its atomic number in terms of x .
- (a) $\frac{4.761}{x}$ (b) $\frac{0.529}{x}$ (c) $\frac{1587}{x}$ (d) $\frac{2.116}{x}$
10. The radius of the fourth orbit of single electron species is 2.116 A° . Calculate the energy of the orbit.
- (a) -27.2 eV (b) -13.6 eV
(c) -40.8 eV (d) -54.4 eV
11. Arrange the elements 'P', 'Q', 'R' and 'S' in the descending order of their atomic numbers based on the following description.
- (A) Element P has 1 electron in N shell, and the number of electrons in the penultimate shell is more than 8 but less than 18.
(B) Element Q has 4 electrons in N shell.
(C) Element R has 1 electron in N shell and the ratio of electrons in K, L, and M shells is 1:4:4.
(D) Element S is an unreactive gas with 4 shells.
- (a) BCDA (b) CDBD (c) ABDC (d) DBAC
12. Arrange the elements 'P', 'Q', 'R' and 'S' in the descending order number of electrons in the outermost shell.
- (a) The number of electrons in the third shell of element P is 9.
(b) The element Q has 2 electrons less than P.
(c) The element R has 2 electrons less than Q.
(d) The element S has 10 electrons more than P.
- (a) BCDA (b) CDAB (c) ACBD (d) DCBA
13. **Assertion (A):** An ion is more stable than an atom.
Reason (R): An ion is a charged particle.
- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.
14. **Assertion(A):** The specific charge of α particle is less than H^+ .
Reason(R): Hydrogen has only one proton.
- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. Match the values of Column A with those of Column B.

Column A (Change in angular momentum of electron during transition in a hydrogen atom)	Column B (Number of spectral lines)
(i) $\frac{5h}{2\pi}$ to $\frac{h}{\pi}$	(A) 1
(ii) $\frac{3h}{2\pi}$ to $\frac{h}{\pi}$	(B) 3
(iii) $\frac{3h}{\pi}$ to $\frac{h}{2\pi}$	(C) 6
(iv) $\frac{2h}{\pi}$ to $\frac{h}{\pi}$	(D) 15

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
 (b) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (D)
 (c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
 (d) (i) $\rightarrow$ (A); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)

Space for rough work

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Arrange the elements corresponding to the following atomic numbers in the decreasing order of valence electrons.

(A) 15 (B) 30 (C) 14 (D) 35

(E) 31

(a) BCEAD

(b) DCEAB

(c) BAECD

(d) DACEB

2. Radius and energy of Bohr's orbits in hydrogen atom are given below. Arrange the values in the decreasing order of ' n ' values.

(A) 2.116 Å

(B) 0.53 Å

(C) 8.464 Å

(D) -1.51 eV

(a) DBAC

(b) DACB

(c) CDAB

(d) CBDA

3. **Assertion (A):** The source of α -particles is taken in a lead block.

Reason (R): Some radioactive emanations can penetrate through lead block.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true and R is false.

(d) A is false and R is true.

4. **Assertion (A):** Potential energy of an electron increases with increase in the distance from the nucleus.

Reason (R): Velocity of an electron decreases with increase in the distance from the nucleus.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true and R is false.

(d) A is false and R is true.

5. An electron in hydrogen atom is excited to the sixth orbit. It comes back to the second energy level during de-excitation. Identify the difference in angular momentums experienced by the electron.

(a) $\frac{h}{2\pi}$

(b) $\frac{2h}{\pi}$

(c) $\frac{h}{\pi}$

(d) $\frac{h}{4\pi}$

6. A dinegatively charged ion has same number of protons and neutrons. Its mass number is 32. The number of electrons in the ion is _____.

(a) 18

(b) 16

(c) 14

(d) 20

7. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Presence of central nucleus	(A) Thomson's atomic model
(ii) Presence of circular orbits	(B) Dalton's atomic theory
(iii) Atoms can neither be created nor be destroyed	(C) Planck's quantum theory
(iv) Negatively charged particles embedded in positively charged mass	(D) Bohr's atomic model
(v) Quantization of energy	(E) Rutherford's atomic model

- (a) (i) → (C); (ii) → (A); (iii) → (E); (iv) → (B); (v) → (D)
 (b) (i) → (D); (ii) → (E); (iii) → (B); (iv) → (A); (v) → (D)
 (c) (i) → (E); (ii) → (D); (iii) → (B); (iv) → (A); (v) → (C)
 (d) (i) → (E); (ii) → (A); (iii) → (D); (iv) → (B); (v) → (C)

8. Atom of an element 'X' has 'N' shell as valence shell. The sum of electrons in K, L and N shells is equal to the number of electrons in 'M' shell. Find the atomic number of X.

- (a) 20 (b) 18 (c) 36 (d) 38

9. The mass of protons is 36,740 times that of an electron. Identify the false statement regarding the element.

- (a) The element can form a dipositive ion.
 (b) 'N' shell is the valence shell in the element.
 (c) The ion formed has 18 electrons in the outermost shell.
 (d) The atom of this element has 18 core electrons.

10. The amount of energy required to remove an electron from a particular orbit of hydrogen atom is x Joules. What is the amount of energy required to remove the electron from the same orbit of He^+ ion?

- (a) $2x$ J (b) $\frac{x}{4}$ J (c) $4x$ J (d) $\frac{x}{2}$ J

11. In Millikan's oil drop experiment, the charges of three oil drops 'A', 'B' and 'C' are 1.2 C, 2.8 C and 0.72 C, respectively. What would be the number of electrons attached to these oil drops?

- (a) 3, 7, 9 (b) 15, 35, 9
 (c) 30, 70, 18 (d) 8, 16, 24

12. Oxygen exists in three isotopic forms with mass numbers 16, 17 and 18. The percentage abundances of these isotopes are 81%, 9% and 10%, respectively. Calculate the average atomic mass of the element.

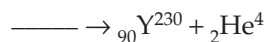
- (a) 16.9 (b) 17.2 (c) 16.29 (d) 16.5

Space for rough work

13. Hypothetical values of charge of electron, masses of proton and neutron are 3.2×10^{-19} C, 8.35×10^{-28} g and 8.6×10^{-28} g, respectively. Calculate the specific charge of an α -particle.

- (a) 5.6×10^9 C/kg (b) 3.8×10^8 C/kg
(c) 1.9×10^8 C/kg (d) 0.95×10^9 C/kg

14. Complete the following reaction:



- (a) ${}_{91}\text{X}^{231}$ (b) ${}_{88}\text{X}^{226}$ (c) ${}_{92}\text{X}^{230}$ (d) ${}_{92}\text{X}^{234}$

15. Which of the following conditions is most suitable to be maintained in the discharge tube?

- (a) High voltage, high pressure
(b) High voltage, low pressure
(c) Low voltage, low pressure
(d) Low voltage, high pressure

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Arrange the elements corresponding to the following atomic numbers in the increasing order of number of electrons in 'M' shell.

(A) 26 (B) 22 (C) 16 (D) 28
(E) 33

(a) CBADE (b) CDABE
(c) EBADC (d) EDABC

2. Arrange the following sets of orbits and species in the increasing order of energies.

(A) H atom, $n = 2$ (B) He^+ ion, $n = 2$
(C) Li^{+2} ion, $n = 2$ (D) Be^{+3} ion, $n = 2$
(a) ADCB (b) DCBA (c) ABCD (d) DBAC

3. **Assertion (A):** The lead container in which the source of α -particles is kept has perforation on one side.

Reason (R): α -particles produce scintillations on the ZnS screen.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

4. **Assertion (A):** Energy of an electron in the second orbit is greater than the energy of electron in the first orbit.

Reason (R): Both potential energy and kinetic energy of electron increase with increase in the distance from the nucleus.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

5. Identify the angular momentum which is not possible for an electron in Bohr's orbit of a hydrogen atom.

(a) $\frac{h}{\pi}$ (b) $\frac{3h}{2\pi}$ (c) $\frac{4h}{\pi}$ (d) $\frac{3h}{4\pi}$

6. Mass number of an element is double the atomic number. There are 26 neutrons in the atom of the element. The number of electrons present in the penultimate shell of dipositive ion is _____.

(a) 18 (b) 8 (c) 14 (d) 13

7. Match the entries of Column A with those of Column B.

Column A	Column B
(i) $E = \frac{hc}{\lambda}$	(A) Rutherford
(ii) $mvr = \frac{nh}{2\pi}$	(B) J. J. Thomson
(iii) α -ray scattering experiment	(C) Max Planck
(iv) Thinly spread positively charged mass	(D) Niels Bohr

(a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

(b) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)

(c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)

(d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)

8. The sum of electrons present in the second and fourth shells is equal to the number of electrons in the third shell of an element and the fifth shell is valence shell. Find out the number of core electrons in B.

(a) 18 (b) 19 (c) 38 (d) 36

9. A dinegative ion of an element has protons whose mass is equal to 14,696 times that of an electron. The number of neutrons is more than the number of protons by 2. What could be the mass number?

(a) 16 (b) 8 (c) 18 (d) 10

10. The same amount of energy is required to remove an electron from the respective orbits of hydrogen atom and He^+ ion. Identify the orbits.

(a) $n = 2; n = 1$ (b) $n = 2; n = 3$

(c) $n = 2; n = 4$ (d) $n = 4; n = 1$

11. In Millikan's oil drop experiment, the charges on the oil drops A, B and C are 0.3 C, 0.12 C and 0.18 C, respectively. What would be the number of electrons attached to these oil drops?

(a) 3, 12, 18 (b) 6, 24, 38

(c) 5, 2, 3 (d) 6, 4, 9

12. An element exists in the form of two isotopes with mass numbers x and $x + 2$. If the percentage abundance of these isotopes is 75.77% and 24.23%, then what is the average atomic mass of the element?

(a) $x + 0.5$ (b) $x + 0.25$

(c) $x + 0.75$ (d) $x + 1$

13. Deuterium is an isotope of hydrogen. Calculate the specific charge of deuteron (charge of electron = 1.6×10^{-19} C; mass of proton = 1.67×10^{-27} kg; mass of neutron = 1.72×10^{-27} kg)

(a) 2.3×10^7 C/kg (b) 1.15×10^7 C/kg

(c) 0.575×10^8 C/kg (d) 3.45×10^8 C/kg

Space for rough work

14. Which of the following changes occur due to the emission of α -particle from an atom of an element?
- Atomic number and mass number increase by 2 units and 4 units, respectively.
 - Atomic number and mass number decrease by 2 units and 4 units, respectively.
 - Only atomic number increases by one unit.
 - Only atomic number decreases by one unit.
15. In discharge tube, why are high voltage and low pressure required to be maintained?
- In order to increase the kinetic energy of the electron dislodged from the atom
 - In order to increase the number of dislodged electrons
 - In order to make the fluorescent screen glow properly
 - All the above

Answer Keys

Assessment Test I

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (b) | 3. (b) | 4. (b) | 5. (a) | 6. (d) | 7. (d) | 8. (b) | 9. (a) | 10. (b) |
| 11. (c) | 12. (b) | 13. (c) | 14. (a) | 15. (d) | | | | | |

Assessment Test II

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (a) | 3. (b) | 4. (b) | 5. (b) | 6. (d) | 7. (c) | 8. (c) | 9. (a) | 10. (b) |
| 11. (d) | 12. (b) | 13. (b) | 14. (a) | 15. (a) | | | | | |

Assessment Test III

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (d) | 2. (c) | 3. (a) | 4. (b) | 5. (b) | 6. (a) | 7. (c) | 8. (c) | 9. (c) | 10. (c) |
| 11. (b) | 12. (c) | 13. (c) | 14. (d) | 15. (b) | | | | | |

Assessment Test IV

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (b) | 3. (b) | 4. (c) | 5. (d) | 6. (b) | 7. (a) | 8. (c) | 9. (c) | 10. (c) |
| 11. (c) | 12. (a) | 13. (b) | 14. (b) | 15. (d) | | | | | |